SIMATS ENGINEERING
ASSESSMENT TOOL 30

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

RUBRICS FOR CALCULATION:

Annexure A
Scenario based Questions

Q1. Data Cleaning and Normalization in Healthcare Analytics (L3, L4, L5)

A healthcare analytics company is developing a model to predict patient readmission. The dataset collected from different hospitals contains missing values, inconsistent categorical entries, and extreme values.

Sample Dataset

Patient_ID	Age	Gender	BMI	Annual_Visits	Blood_Pressure	Risk_Level
P1	25	M	22.5	3	120	Low
P2	NaN	Male	24.8	5	130	Medium
P3	42	F	NaN	7	135	Medium
P4	110	Female	45.2	50	200	High
P5	36	female	29.5	NaN	140	Medium
P6	50	FEMALE	31.2	12	150	High
P7	29	male	21.8	4	125	Low
P8	45	Female	60.0	40	190	High

Tasks

A. Missing Value Treatment

i. Identify missing values in Age, BMI, and Annual_Visits.

ii. Apply Median Imputation and Mean Imputation. iii. Compare the resulting values and justify which method is more appropriate.

B. Outlier Detection

i. Detect outliers in Age and Annual_Visits using the Z-score method.

ii. Identify the records containing extreme observations. iii. Apply capping or removal and justify the selected approach.

C. Categorical Standardization

i. Identify inconsistent Gender values. ii. Convert all values into Male/Female. iii. Explain why categorical standardization is important for machine-learning models.

D. Normalization

i. Apply Min-Max normalization to BMI.

ii. Explain the effect of normalization on model performance.

Q2. Covariance and Feature Selection in Banking Analytics (L4, L5)

A bank wants to develop a customer-risk prediction model. Several attributes are strongly related, resulting in redundant information and increased computational complexity. Sample Dataset

Customer_ID	Age	Income (₹K)	Loan_Amount (₹K)	Credit_Score	Monthly_EMI (₹)	Savings (₹K)	Risk
B1	25	35	100	650	5000	20	Low
B2	32	50	150	680	7000	30	Low
B3	40	70	200	720	9000	45	Medium
B4	48	100	300	780	14000	70	High
B5	29	42	120	660	5500	25	Low
B6	38	65	180	710	8500	40	Medium
B7	55	130	350	820	17000	90	High
B8	44	85	250	760	12000	60	High

Tasks

A. Covariance Analysis

i. Calculate the covariance between Income and Loan_Amount. ii.
Determine whether the relationship is positive or negative. iii.
Explain what a high positive covariance indicates in this dataset.

B. Feature Relationship Analysis

i. Identify pairs of attributes that may be highly correlated.

ii. Explain why Age, Income, Savings, and Loan_Amount may contain redundant information. iii. Identify at least three features that could be removed.

C. Feature Selection

i. Select an optimal subset of features for predicting Risk.

ii. Justify your selection using correlation and domain knowledge. iii. Explain how feature selection reduces computational cost.

D. Standardization

i. Standardize all numerical attributes using Z-score normalization.

ii. Explain why Credit_Score and Income cannot be directly compared without scaling.

Q3. Discretization of Sales Data in Retail Analytics (L3, L4, L5)

A retail company wants to classify customers based on the amount they spend per month. Continuous spending values are difficult for managers to interpret, so the company plans to use discretization. Sample Dataset

Customer_ID	Monthly_Spend (₹)	Transactions	Items_Bought	Customer_Category
C1	500	4	6	Basic
C2	750	5	9	Basic
C3	1000	7	12	Basic
C4	1500	9	18	Silver
C5	2000	12	25	Silver
C6	2500	14	30	Gold
C7	3000	16	36	Gold
C8	4000	20	45	Gold
C9	5000	25	55	Platinum
C10	6500	30	70	Platinum

Tasks

A. Equal-Width Binning

i. Divide Monthly_Spend into 4 equal-width bins.

ii. Calculate the bin width. iii. Define the intervals and assign each customer to a bin.

B. Equal-Frequency Binning

i. Divide the dataset into 4 bins containing approximately equal numbers of records.

ii. Assign each customer to the corresponding bin. iii. Compare equal-width and equal-frequency results.

C. Concept Hierarchy

Construct a concept hierarchy:

Monthly_Spend → Low → Medium → High → Very High

D. Interpretation

i. Explain how discretization improves decision-making.

ii. Discuss one disadvantage of converting continuous values into categories.

Q4. Data Transformation in Manufacturing Quality Analytics (L3, L4, L5)

A manufacturing company records machine operating temperatures. The measurements vary considerably, and some machines produce unusually high readings.

Sample Dataset

Machine_ID	Temperature (°C)	Operating_Hours	Vibration
M1	55	100	2.1

M2	60	120	2.5
M3	62	140	2.8
M4	65	160	3.0
M5	70	180	3.2
M6	75	200	3.8
M7	80	220	4.1
M8	120	250	8.5

Tasks

A. Min-Max Normalization

i. Normalize Temperature into the range [0,1].

ii. Calculate the normalized value for each machine. iii. Identify which observation obtains the maximum normalized value.

B. Z-Score Normalization

i. Calculate the mean Temperature. ii. Calculate the standard deviation. iii. Compute the Z-score for each temperature value.

C. Outlier Analysis

i. Identify whether 120°C can be considered an extreme value using the Z-score method.

ii. Explain how this value affects mean-based normalization. iii. Suggest an appropriate treatment method.

D. Comparison

Compare Min-Max and Z-score normalization in terms of their sensitivity to extreme observations.

Q5. Missing Value Imputation and Data Transformation in Student Analytics (L3, L4, L5)

An educational institution is building a student-performance prediction model. Data collected from different departments contain missing examination marks, attendance values, and study hours.

Sample Dataset

Student_ID	Study_Hours	Internal_Mark	Attendance (%)	Assignment_Score	Grade
S1	2	55	70	60	C
S2	3	NaN	75	65	C
S3	4	68	NaN	72	B
S4	5	75	85	78	B
S5	NaN	80	90	82	A
S6	7	88	92	NaN	A
S7	8	92	95	90	A
S8	12	98	98	95	A

Tasks
A. Missing Value Analysis
i. Identify all missing observations.
ii. Apply mean imputation and median imputation. iii. Determine which method is more suitable for Study_Hours and explain why.
B. Data Transformation
i. Apply Min-Max normalization to Internal_Mark.
ii. Apply Z-score normalization to Assignment_Score after imputing the missing value. iii. Compare the transformed results.
C. Discretization
Divide Study_Hours into three categories:

- Low
- Medium
- High

Define suitable thresholds and assign each student to a category.
D. Interpretation
Explain how data preprocessing can improve the accuracy and interpretability of a student-performance prediction model.

Solutions :

Q1. Data Cleaning and Normalization in Healthcare Analytics

A. Missing Value Treatment

Given Dataset (Identified Missing Values)

Patient	**Age**	**BMI**	**Annual Visits**
P1	NaN	22.5	3
P2	42	24.8	5
P3	NaN	NaN	7
P4	36	29.5	NaN
P5	50	31.2	12
P6	29	21.8	4
P7	45	60.0	40

Missing values:

- Age: P1, P3
- BMI: P3
- Annual Visits: P4

Mean Imputation

Age Mean

Available ages:

42, 36, 50, 29, 45

$$Mean = \frac{42 + 36 + 50 + 29 + 45}{5} = 40.4$$

Missing Age values replaced by **40.4**

BMI Mean

$$Mean = \frac{22.5 + 24.8 + 29.5 + 31.2 + 21.8 + 60}{6} = 31.63$$

Missing BMI replaced by **31.63**

Annual Visits Mean

$$Mean = \frac{3 + 5 + 7 + 12 + 4 + 40}{6} = 11.83$$

Missing Annual Visits replaced by **11.83**

Median Imputation

Median values:

Age:

29,36,42,45,50

Median = **42**

BMI:

21.8,22.5,24.8,29.5,31.2,60

Median = (24.8+29.5)/2 = **27.15**

Annual Visits:

3,4,5,7,12,40

Median = (5+7)/2 = **6**

Comparison

Attribute	Mean	Median
Age	40.4	42
BMI	31.63	27.15
Visits	11.83	6

Decision: Median imputation is more suitable because healthcare data contains extreme values (BMI 60 and visits 40). Median is less affected by outliers.

B. Outlier Detection Using Z-score

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Extreme values usually have:

$$|Z| > 3$$

Annual visits contain:

40 visits

This value is much higher compared with other patients, so it is an outlier.

BMI:

60 is also an extreme BMI value.

Treatment

Use **capping method** instead of removal because healthcare records are important.

Capping replaces extreme values with upper limit values.

C. Categorical Standardization

Original values:

Male, F, Female, female, FEMALE, male

Converted values:

Original	Standardized
Male	Male
F	Female
Female	Female

female	Female
FEMALE	Female
male	Male

Importance

Machine learning algorithms cannot directly process different labels. Standardization avoids duplicate categories and improves prediction accuracy.

D. Min-Max Normalization of BMI

Formula:

$$X' = \frac{X - Min}{Max - Min}$$

Minimum BMI = 21.8

Maximum BMI = 60

Range:

$$60 - 21.8 = 38.2$$

For BMI = 22.5:

$$\frac{22.5 - 21.8}{38.2} = 0.018$$

For BMI = 60:

$$\frac{60 - 21.8}{38.2} = 1$$

Effect

Normalization converts values into 0–1 range. It improves model performance by preventing attributes with large values from dominating.

Q2. Covariance and Feature Selection in Banking Analytics

A. Covariance between Income and Loan Amount

Income values:

35,50,70,100,42,65,130,85

Loan Amount:

100,150,200,300,120,180,350,250

Mean Income:

$$\bar{x} = 72.125$$

Mean Loan:

$$\bar{y} = 206.25$$

Covariance calculation gives:

$$Cov(x,y) = 2192.18$$

Result

Covariance is positive.

Interpretation

Customers with higher income generally receive higher loan amounts.

B. Feature Relationship Analysis

Highly related attributes:

- Income and Loan Amount
- Income and Savings
- Age and Income

Redundant features:

- Income
- Savings
- Loan Amount

because they represent financial capacity.

C. Feature Selection

Selected features:

1. Credit Score
2. Income

3. Age
4. Loan Amount

Removed:

- Savings (similar information to income)
- Monthly EMI (depends on loan amount)

Advantage

Feature selection reduces:

- computation time,
- storage requirement,
- model complexity.

D. Standardization

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Z-score converts different scales into a common range.

Example:

Income is in thousands, while Credit Score is around 700.

Without scaling, Income may dominate the model.

Q3. Discretization of Sales Data

A. Equal Width Binning

Monthly Spend:

Minimum = 500

Maximum = 6500

Range:

$$6500 - 500 = 6000$$

Number of bins = 4

Bin width:

$$6000/4 = 1500$$

Bins:

Bin	**Range**
Low	500-2000
Medium	2000-3500
High	3500-5000
Very High	5000-6500

B. Equal Frequency Binning

10 records / 4 bins ≈ 2–3 records each.

Bins:

Bin 1:

500,750,1000

Bin 2:

1500,2000,2500

Bin 3:

3000,4000

Bin 4:

5000,6500

Comparison

Equal width gives fixed intervals.

Equal frequency gives balanced number of records.

C. Concept Hierarchy

Monthly Spend →

Low → ₹500–2000

Medium → ₹2000–3500

High → ₹3500–5000

Very High → Above ₹5000

D. Interpretation

Discretization converts continuous spending into understandable categories.

Advantages:

- easier customer classification,
- better decision making.

Disadvantage:

- loss of exact information.

Q4. Data Transformation in Manufacturing Analytics

A. Min-Max Normalization

Formula:

$$X' = \frac{X - Min}{Max - Min}$$

Minimum temperature = 55

Maximum temperature =120

Range:

$$65$$

For 55°C:

$$\frac{55 - 55}{65} = 0$$

For 120°C:

$$\frac{120 - 55}{65} = 1$$

Normalized values:

Temperature	Normalized
55	0
60	0.077
62	0.108
65	0.154
70	0.231
75	0.308
80	0.385

120	1

Maximum normalized value = **120°C**

B. Z-score Normalization

Mean:

$$\mu = \frac{55 + 60 + 62 + 65 + 70 + 75 + 80 + 120}{8}$$

$$\mu = 73.375$$

Standard deviation ≈ 21.16

Example:

For 55°C:

$$Z = \frac{55 - 73.375}{21.16}$$

$$Z = -0.86$$

C. Outlier Analysis

120°C has a high positive Z-score.

It can be considered an outlier.

Treatment:

- verify machine condition,
- apply capping,
- or remove if it is measurement error.

D. Comparison

Min-Max	**Z-score**
Sensitive to outliers	Less sensitive
Range 0-1	Mean centered
Good for bounded data	Good for normal distributions

Q5. Student Analytics

A. Missing Value Analysis

Missing values:

- Study Hours: S5
- Attendance: S3
- Assignment Score: S7

Mean Imputation

Study Hours:

$$Mean = \frac{2+3+4+5+7+8+12}{7} = 5.85$$

Replace missing value with 5.85.

Median Imputation

Sorted:

2,3,4,5,7,8,12

Median = 5

Replace missing value with 5.

Decision

Median is better because study hours may contain extreme values.

B. Data Transformation

Min-Max Normalization

Formula:

$$X' = \frac{X - Min}{Max - Min}$$

It converts Internal Marks into 0–1 scale.

Z-score Normalization

Assignment score is converted using:

$$Z = \frac{x - \mu}{\sigma}$$

This shows how far a student's score is from average performance.

C. Discretization of Study Hours

Categories:

Hours	Category
0-4	Low
5-8	Medium
Above 8	High

Example:

S1 (2 hours) → Low

S5 (7 hours) → Medium

S7 (12 hours) → High

D. Interpretation

Data preprocessing improves student prediction models by:

- handling missing data,
- reducing noise,
- scaling attributes,
- improving model accuracy,
- making results easier to interpret.